



CLOUD COMPUTING — MINI PROJECT

Azure File Sync with an On-Premises File Server

Extending a local Windows file server into the cloud — centralized storage, multi-site sync and cloud tiering using Microsoft Azure.

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AGENDA

What we'll cover today

01

The Problem

Why a single, local file server isn't enough anymore

03

Architecture

How the on-prem server, agent and Azure talk to each other

05

Setting It Up

Step-by-step from resource group to a synced folder

02

What is Azure File Sync

The concept, and how it fits with an existing server

04

Azure Services Used

Every resource the project depends on, and why

06

Results & Takeaways

What worked, what we'd improve, and the outcome

PROBLEM STATEMENT

A local file server alone doesn't scale

Most organisations still run a Windows file server on-site because staff need fast, familiar drive-letter access. But that server has fixed disks, one location, and no built-in disaster recovery — every GB of growth means buying more storage.



Storage keeps running out

Local disks fill up; scaling means new hardware, every time.



No access for other branches

A second office or remote worker can't reach the same files easily.



Single point of failure

If the server or its disk fails, there is no automatic offsite copy.



Manual backup & DR

Backups are scripted, on tape/NAS, and rarely tested end-to-end.

THE CONCEPT

What is Azure File Sync?

Azure File Sync is a service that keeps a folder on your on-premises Windows Server in sync with an Azure file share in the cloud. The Azure share becomes the central copy; the local server keeps a fast, cached copy — so users still open files over the normal network drive, but the data is now backed by Azure.



Multi-site sync — every registered server sees the same files.



Cloud tiering — cold files move to Azure, freeing local disk.



Built-in disaster recovery — data is safe even if the server dies.

HOW THE DATA FLOWS



On-Premises File Server

Windows Server + shared folder, used daily by staff

↓ Azure File Sync Agent



Storage Sync Service

Coordinates the sync group & endpoints in Azure

↓ Keeps both copies identical

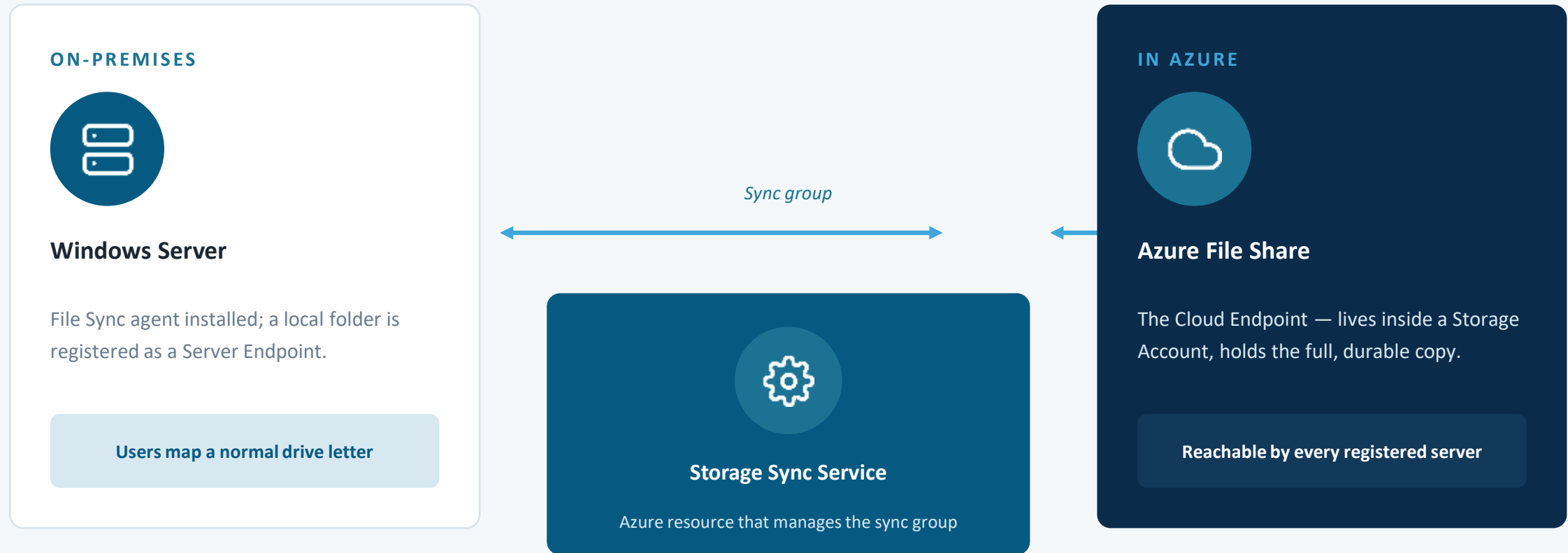


Azure File Share

The durable, central cloud copy of every file

How the pieces connect

One sync group ties the on-prem server and the Azure file share together; either side can change a file and the other catches up.



Azure services this project uses



Storage Account

The parent resource that holds the file share and all its data.



Azure File Share

The actual cloud folder — an SMB share reachable from anywhere.



Storage Sync Service

The regional Azure resource that manages sync groups.



Sync Group

Links one cloud endpoint to one or more server endpoints.



Server Endpoint

The on-prem folder path being kept in sync (via the agent).



Azure AD (Entra ID)

Identity for signing in to the portal and controlling access.



Role-Based Access (IAM)

Restricts who can manage the storage account and its keys.



Azure Backup (optional)

Point-in-time snapshots of the file share for extra safety.

What each service is actually doing

Service	Role in the project	How we used it
Storage Account	Holds the file share, storage keys and access policies	Created once, in the resource group for this project
Azure File Share	Stores the master copy of every project file	Mounted by the Storage Sync Service as the cloud endpoint
Storage Sync Service	Owns the sync group and tracks sync health	One per region; monitors sync status from the portal
Sync Group	Pairs the cloud endpoint with the server endpoint	Any change on either side is detected and pushed across
Server Endpoint / Agent	Runs on the Windows Server, watches the local folder	Installed once; registers the server to the sync service
Azure AD + IAM	Controls who can sign in and manage these resources	Only the project admin role can rotate storage keys

Setting it up, step by step

01

Create a Storage Account

Resource group → Storage Account → choose region and redundancy (LRS).

02

Add an Azure File Share

Inside the storage account, create a file share and set its quota.

03

Deploy Storage Sync Service

Create the Storage Sync Service resource in the same region.

04

Create a Sync Group

Link the sync group to the file share — this becomes the cloud endpoint.

05

Install the agent on-prem

Run the Azure File Sync agent installer on the Windows Server.

06

Register server & add endpoint

Register the server to the sync service, then add the local folder as a server endpoint.

What it looks like in the portal

Simplified recreations of the two screens we used most — the sync group overview and the server endpoint status.

portal.azure.com/ storagesyncservice / syncgroups

ProjectSyncGroup — Overview

1 file share

Cloud endpoint

1 registered

Server endpoints

Healthy

Sync status

ENDPOINTS

Cloud endpoint

afs-share-01

● Healthy

Server endpoint

SRV-FILE-01 D:\ProjectData

● Healthy

Last sync: a few seconds ago

portal.azure.com/ serverendpoint / properties

Server Endpoint — Properties

Server

SRV-FILE-01

Registered path

D:\ProjectData

Cloud tiering

Enabled — 20% free space

Provisioning state

Succeeded

Health

Healthy

Files synced

1,248 items

OUTCOME

What this setup actually gave us



Freed-up local disk

Cloud tiering moved cold files to Azure automatically — the same server disk now holds far more than its physical size.



Real disaster recovery

The Azure file share is the durable copy; even a full server crash doesn't mean lost data.



Same files, any site

A second server (or a new one) can join the same sync group and get every file automatically.



No change for end users

Staff still use a normal mapped drive — the sync happens in the background, nothing to relearn.

REFLECTION

Challenges we ran into

1

Initial sync took a while

The first sync has to walk every file, so a large folder took noticeably longer than later, incremental syncs.

2

Firewall / outbound access

The agent needs outbound HTTPS to Azure; a locked-down network needed a couple of endpoints allow-listed first.

3

Understanding cloud tiering

It took some testing to see how 'free space policy' decides which files stay local versus move to the cloud.

4

Keeping storage keys safe

Storage account keys give full access — we moved to role-based access for day-to-day use instead of sharing keys.

CONCLUSION

A hybrid file server that just works

Azure File Sync let us keep the on-premises server our users already know, while Azure quietly took over storage growth, backup and multi-site access in the background.

- ✓ Local performance, cloud durability — best of both.
- ✓ Scales by editing a quota, not buying new disks.
- ✓ A pattern real organisations use for hybrid storage today.

Thank you

Questions welcome